BOXIANG ZHANG

West Lafayette, IN | zhan4653@purdue.edu | +1-951-288-8754
linkedin.com/in/boxiang-zhang-369086201

EDUCATION

Ph.D. SOET: Machine Learning and Data Science *Aug 2022 - Aug 2026 (Expected)*
Purdue University, West Lafayette, IN

M.S. in ECE (with Honors): Machine Learning and Data Science *Aug 2020 - May 2022*
University of Southern California, Los Angeles, CA

B.E. EE: Optical and Electronic information *Sep 2019 - Jun 2020*
Huazhong University of Science and Technology

SKILLS

Machine Learning & AI Expertise: Deep Learning, Computer Vision, Transformers, Diffusion Models, Anomaly Detection, Multi-Modal Learning, Self-Supervised Learning, 3D Object Detection, Foundation Models, Large Language Models (LLMs), Normalizing Flows, Optimal Brain Surgery (OBS)

Technical Skills & Tools: End-To-End development, Python, PyTorch, TensorFlow, Scikit-learn, OpenCV, vLLM, ROS, Spark, Pandas, SQL, MATLAB, JavaScript, Bash, GPU Computing, Model Training & Deployment, Image Acquisition Systems, Data Pipeline Design, Git/GitHub, LaTeX, Linux, Ubuntu, Shell, Data Visualization, Edge Computing, Docker, Statistics

Communication & Teaching: Technical Communication, Mentorship & Knowledge Transfer, Workshop Instruction, Cross-Disciplinary Collaboration, Team-working, Problem-Solving

RESEARCH EXPERIENCE

CORP: Closed-Form One-shot Representation-Preserving Structured Pruning for Vision Transformers *Dec 2025 - Present*
Purdue University | West Lafayette, IN

- Developed a one-shot structured pruning method for Vision Transformers without retraining, using small unlabeled calibration sets.
- Introduced closed-form compensation to correct structured pruning bias and preserve intermediate representations.
- Maintained 82.8% Top-1 accuracy on DeiT-Huge at 50% sparsity, achieving ~1.9× throughput improvement on GPU hardware.
- Designed scalable post-training compression pipeline suitable for deployment-constrained environments.

Efficient Transformer Compression *Aug 2024 - Present*
Purdue University | West Lafayette, IN

- Designed a structured one-shot pruning method for pre-trained Vision Transformers without retraining, using a small dataset (4K ImageNet images).
- Applied Optimal Brain Surgery (OBS) with masked structured pruning, leveraging the Woodbury identity to efficiently approximate the Hessian inverse and accelerate parameter updates.
- Reduced model redundancy by pruning 50% of parameters in MLP modules while maintaining high task accuracy (ImageNet validation: 81.8% → 76.7%) leading to a more efficient model with minimal accuracy loss.

High-Resolution Depth Prediction Using RGB Images and LiDAR-Guided Training *Mar 2025 - Present*
Purdue University | West Lafayette, IN

- Integrated depth estimation models for drone imagery under limited training data conditions.
- Developed a method to generate high-resolution ground truth from low-resolution LiDAR data to fine-tune depth estimation models.
- Achieved high-resolution depth predictions with high accuracy ($R^2 > 0.9$), demonstrating effective supervision transfer and strong generalization.

Digital Forestry via Multi-Modal Learning *Sep 2024 - Feb 2025*
Purdue University | West Lafayette, IN

- Developed a multi-modal machine learning pipeline combining aerial imagery and language–vision models for forest segmentation and environmental reasoning.
- Enhanced the Segment Anything Model (SAM) with crown-top features, increasing crown detection from ~40 to 200+ crowns per image to enable finer-grained segmentation.
- Integrated SAM with LLaVA to generate high-level environmental insights for forest monitoring.

AI Vision System for Automated Quality Inspection *Aug 2022 - Jun 2024*
Purdue University | West Lafayette, IN

- Led the development of a real-world AI vision system for manufacturing quality inspection.
- Designed and implemented an anomaly detection pipeline using normalizing flow models to detect subtle defects in high-resolution product images.
- Coordinated entire system, including image acquisition interface and model development.
- The solution achieved 95%+ AUROC in production, enabling accurate and efficient defect detection on the factory floor.

Semantic Error Detection and Correction *Jul 2021 - Feb 2022*
University of Southern California | Los Angeles, CA

- Built a large-scale benchmark dataset for table understanding, pairing structured table data with natural-language descriptions.
- Applied pre-trained word embeddings and an NLI model to cluster and merge semantically related titles, improving label consistency and coverage.
- Enabled scalable evaluation of semantic consistency and reasoning in table tasks, supporting downstream use cases such as LLM hallucination detection and error correction.

LEADERSHIP

Workshop Presenter *Sep 2024 - Present*

Purdue University | West Lafayette, IN

- Developed and delivered 20 interactive workshops (2024–present) reaching 400+ students and early-career professionals, building both foundational and applied knowledge in machine learning.
- Prepared materials on topics including Python and PyTorch programming foundations; deep learning fundamentals; dimensionality reduction and supervision strategies; transformers and large language models (LLMs); diffusion models; model-assisted data generation; transformer compression; anomaly detection; and LLM deployment.
- Designed hands-on examples and simplified complex concepts to support learners across varying experience levels.

PUBLICATIONS

1. B. Zhang, X. Wang, B. Yang, C. Vian, "PatchFlow: Leveraging a Flow-Based Model with Patch Features.", 3rd Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE) 2024
2. B. Zhang, X. Wang, B. Yang, C. Vian, "A.I. Vision System for Automated Casting Quality Inspection.", NADCA Conference and Tabletop 2023, 2023
3. L Jin, B Zhang, M Alqaydi, X Wang, T Taha, D Gan, "Kinetostatic Modeling and Cable Tension Optimization of Cable-Driven Parallel Robots on Cylindrical Surfaces", ASME's IDETC-CIE, 2025
4. L Jin, B Zhang, A AlAttar, M AlQaydi, M Abduljawad, T Taha, D Gan, "Cable Tension Optimization for a Planar CDPR Through Anchor Reconfiguration", ReMAR, 2024